Gyutae Oh

PH.D. RESEARCHER · COMPUTER VISION & CONTINUAL LEARNING

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Ph.D. student at the Media System Lab, Sungkyunkwan University (advisor: Prof. Jitae Shin). I build vision models that stay accurate as conditions change — **domain continual learning** without storing past data, **multimodal learning robust to missing inputs**, and **efficient models** with low compute and memory — across medical, industrial, and natural images. 12 peer-reviewed papers (8 first-author) including WACV 2027 and ACCV 2026 (oral); 5 first-author submissions under review (CVPR ×3, ICLR, IEEE JBHI).

INTERESTS

Domain continual learning · Rehearsal-free / prompt-based CL · Missing-modality learning · Efficient learning & PEFT · Test-time adaptation · Image restoration · Anomaly detection · Medical image analysis

EDUCATION

Ph.D. in Electrical and Computer Engineering, Sungkyunkwan University — Mar 2026 – Feb 2029 (exp.)
Media System Lab · GPA 4.5/4.5 · Thesis (tentative): *Efficient Continual Learning under Distribution Shift*

M.S. in Electrical and Computer Engineering, Sungkyunkwan University — Mar 2024 – Feb 2026
Media System Lab · GPA 4.07/4.5 · Thesis: *Efficient Continual Learning Framework for Medical Imaging AI*

B.S. in Computer Engineering, Korea National University of Transportation — Mar 2019 – Feb 2024
GPA 3.79/4.5

EXPERIENCE

Graduate Researcher, Media System Lab, Sungkyunkwan University — Jan 2024 – Present
Advisor: Prof. Jitae Shin · Continual learning and medical imaging AI

Undergraduate Researcher, Applied Machine Intelligence Lab (AMI-Lab), KNUT — Mar 2022 – Jun 2023
Advisor: Prof. Jeonghwan Gwak · Image and biosignal classification

RESEARCH PROJECTS

UniPrompt-CL: Single-Inference Domain Continual Learning · `WACV 2027, first author` — 2025.09 – 2026.10

- Replaced per-layer prompt pools with one pool shared by all ViT layers, so prompt selection and prediction take a single backbone pass; no past data stored.
- Diabetic retinopathy: **+2.0%p** over the strongest same-backbone method, lowest forgetting (AvgF **0.079**), training cost **~¼** of CODA-Prompt (44.17 GFLOPs).

Multimodal Diagnosis of Exposure Keratopathy & Infectious Keratitis · `with Samsung Medical Center` — 2024.10 – 2026.09

- Led all stages except clinical data collection (4-person team): design, modeling, experiments, papers, KR/PCT patents.
- Contrastive alignment of four illumination modalities into the broad-beam image at training time; single-image inference reaches accuracy **0.841** / F1 **0.838** (**+16%p** vs. single-modality training).
- Single ViT with mask-aware cross-attention for any 1–4 image combination: 3-class accuracy **0.785 → 0.835**, macro AUROC **0.964** (333 patients, 2,413 sets, 10 repeats).

NRF Core Research: Adaptive State Inference from Partially Available Multimodal Data — 2026.09 – (3 yrs)

- ER/ICU deterioration prediction from incomplete ECG, vital-sign, and lab data. One of two core Ph.D. researchers; managing development by five M.S. researchers.

Co-authored research (method advisor) · ACCV 2026 2026.03 – 2026.09

- *Spectral Descriptor Conditioning*: proposed the 2D polar frequency descriptor at the core of the paper — best average PSNR **30.07 dB** over five degradations with only 2.2% extra FLOPs.
- *Drift-adaptive Representation Preservation*: co-designed drift-scaled preservation — **3.0 GB** training memory (4.7–10.7× less) and 13.78 MB storage.

PatchCore++: Noise-Robust Industrial Anomaly Detection · ICCE-Asia 2024, first author 2024.04 – 2024.11

- Feature-level noise augmentation with a Locally Aware Common Patch Maker: BTAD instance AUROC **0.9257 → 0.9347**, +2.7%p on the hardest product.

PUBLICATIONS

12 peer-reviewed papers (8 first-author) · 5 under review · * first / co-first author

International conferences

1. **UniPrompt-CL: Efficient Single-Inference Continual Learning for Medical Image Classification**
   Gyutae Oh*, Jitae Shin
   IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027 · Code
2. **Drift-adaptive Representation Preservation for Domain-Incremental Learning**
   Hyunmin Kong, **Gyutae Oh**, Jitae Shin
   Asian Conference on Computer Vision (ACCV), 2026
3. **Change Detection in Longitudinal Chest X-rays Is Readout-Limited, Not Task-Limited**
   Seungin Baek, **Gyutae Oh**, Jitae Shin
   Asian Conference on Computer Vision (ACCV), 2026 · ORAL
4. **Spectral Descriptor Conditioning: Efficient and Interpretable All-in-One Image Restoration via Polar Frequency Signatures**
   Jungwoo Bae, **Gyutae Oh**, Hyunmin Kong, Jitae Shin
   Asian Conference on Computer Vision (ACCV), 2026
5. **Routing Prompts Across Layers: A Global Memory Module for Prompt-Based Continual Learning**
   Gyutae Oh*, Jitae Shin
   International Conference on Computer Research and Development (ICCRD), January 24, 2026
6. **Multimodal Fusion Framework Using Contrastive Learning for Exposure Keratopathy**
   G Oh*, Y Won, D Lim, J Shin
   International Workshop on Ophthalmic Medical Image Analysis (OMIA), MICCAI 2025 Workshop, LNCS 16209, 2025 · Paper · Code
7. **Multi-Head CNN-Attention Prompt Pools: Strengthening Prompt Combination and Contextual Insight for Domain Adaptation**
   G Oh*, J Shin
   International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), 2025 · Paper
8. **Patch Core++: Augmenting Coreset for Superior Industrial Anomaly Detection**
   G Oh*, J Shin
   IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia), 2024 · Paper
9. **Hyperspectral Image Augmentation Technique for the Classification of Diabetic Retinopathy**
   G Oh*, J Shin
   International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), 2024 · Paper

International journals

1. **Causal Representation-Based Personalized Federated Learning with Causal Graph Consensus for Medical Imaging**
   Wooseok Shin, Zhiqiang Shen, **Gyutae Oh**, Jitae Shin
   Electronics, 15(10), 1983, May 7, 2026 · SCIE · JCR Q2 · Paper

Domestic conferences

1. **Feature Ensemble-based Wolff Parkinson White Syndrome classification through ECG**
   G Oh*, I Kim, B Kim, Y Jeon, J Gwak
   Proceedings of the Korean Society of Computer Information Conference, 2023 · Paper

2. **Identity-CBAM ResNet 기반 얼굴 감정 식별 모듈**
   오규태*, 김인기, 김범준, 곽정환
   한국정보처리학회 학술대회논문집 29 (2), 2022 · Paper

Under review

1. **Broad-Beam-Anchored Auxiliary-Optional Vision Transformer for Screening and Differential Diagnosis of Exposure Keratopathy and Infectious Keratitis**
   Gyutae Oh* et al.
   IEEE Journal of Biomedical and Health Informatics, 2026 · Co-first author · SCIE, JCR Q1
2. **Double-blind submissions (first author): CVPR 2027 (×3) · ICLR 2027**
   Domain continual learning and efficient learning — titles withheld during review

PATENT

Device and Method for Diagnosing Exposure Keratopathy based on Multimodal Fusion Learning 2025

KR 10-2025-0121342 · filed Aug 28, 2025 / **PCT** PCT/KR2026/011329 · filed Aug 10, 2026

Single-image inference reaches **0.841** accuracy and **0.838** F1 on four-level severity grading — an average gain of more than **+16%p** over a single-modality baseline, reducing acquisition burden and improving clinical applicability.

HONORS

Awards

Hanium: Companion Dog Eye Disease Detection Assistant App — Selected Works Prize 2023.11
Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

Companion Dog Eye Disease Detection Assistant App — Silver Prize & Graduation Project 2023.08
Korea National University of Transportation

Hanium: Automated Serving Robot using Object Detection & SLAM — Bronze Prize 2020.08
Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

Campus Contest: Automated Serving Robot using Object Detection & SLAM — Grand Prize (President's Award) 2020.09
Korea National University of Transportation

Scholarships

SKKU BK21 Fellowship (Specialized Scholarship) 2024
Sungkyunkwan University

University Innovation — Merit & Living Expense Scholarship 2024
Sungkyunkwan University

Academic Excellence Scholarship (awarded 5 times) 2019–2023
Korea National University of Transportation

Service & certification

Reviewer 2026
IEEE International Conference on Image Processing (ICIP)

Advanced Data Analytics Semi-Professional (ADsP) 2025
Issued by Korea Data Agency (KDATA) · Sep 2025

SKILLS

Core	Python, PyTorch, torchvision, timm, NumPy; Linux multi-GPU experimentation
Libraries	Hugging Face Transformers & PEFT, Ultralytics YOLO, scikit-learn, SciPy, Pandas, Albumentations
Efficiency	Prompt / prefix tuning, knowledge distillation, FLOPs and GPU-memory profiling
Workflow	Git, TensorBoard, Weights & Biases, LaTeX; multi-seed evaluation with confidence intervals
Also used	C/C++, Java, TensorFlow/Keras, ONNX, TensorRT, TFLite

MILITARY

Republic of Korea Army — Sergeant, honorably discharged 2020 – 2022
TOD operator, Gachilbong GOP, 21st Infantry Division (III Corps)